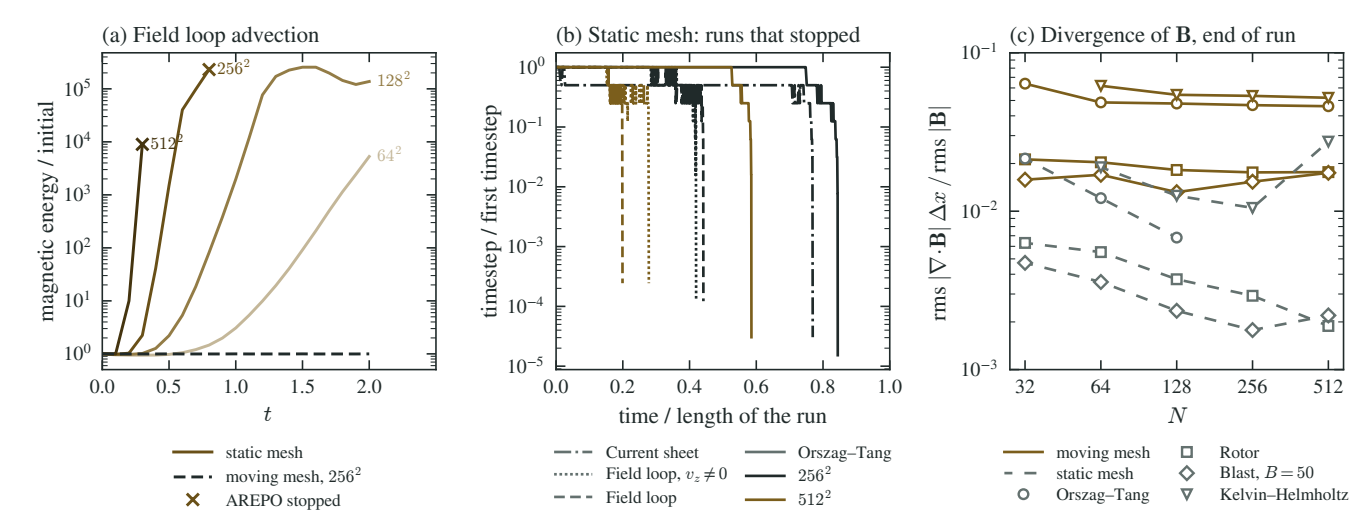
(a) Field loop advection
magnetic energy / initial
$10^0$
$10^1$
$10^2$
$10^3$
$10^4$
$10^5$
0.0
0.5
1.0
1.5
2.0
$t$
$256^2$
$512^2$
$128^2$
$64^2$
static mesh
moving mesh, $256^2$
AREPO stopped
(b) Static mesh: runs that stopped
timestep / first timestep
$10^0$
$10^{-1}$
$10^{-2}$
$10^{-3}$
$10^{-4}$
$10^{-5}$
0.0
0.2
0.4
0.6
0.8
1.0
time / length of the run
Current sheet
Field loop, $v_z \neq 0$
Field loop
Orszag–Tang
$256^2$
$512^2$
(c) Divergence of B, end of run
rms $|\nabla \cdot \mathbf{B}|\,\Delta x$ / rms $|\mathbf{B}|$
$10^{-1}$
$10^{-2}$
$10^{-3}$
32
64
128
256
512
$N$
moving mesh
static mesh
Orszag–Tang
Rotor
Blast, $B = 50$
Kelvin–Helmholtz